
When Do Aligned Representations Agree on What They Do Not Know?

CKA, Prior Strength and the Spectral Tail

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Abstract

Representational alignment, measured by mutual nearest neighbours in the platonic-representation literature and by centred kernel alignment (CKA) more broadly, is the main evidence that different vision models learn similar representations. We ask when linear heads on two encoders agree about *what they do not know*, and show that high CKA does not imply it, for two reasons we make exact. For Bayesian linear heads on Gaussian features, the correlation across inputs of two heads’ epistemic variances equals a ridge-regularised alignment index S_ρ at the heads’ own prior strength. CKA is the $\rho \rightarrow \infty$ limit of this index and weights spectral directions by squared variance, whereas epistemic uncertainty (EU) counts every direction above ρ ; we construct pairs with CKA > 0.99 and EU correlation 0.06, and the converse. CKA is also blind to feature scale, which acts on EU as a change of prior; so are whitened indices, and only S_ρ evaluated at the head’s prior strength on the head’s own feature scaling is not. Representation-only non-identification follows already from the dependence of EU on prior and data; our results say *which* spectral information an index needs and predict that whitened, canonical-correlation-type indices should outperform CKA for weakly regularised heads. On five frozen encoders with CIFAR-10H labels, the evidence is consistent with this but limited: across 30 encoder pairs, S_ρ , its CCA-type limit and linear predictivity order confidence-controlled EU agreement at Spearman ≈ 0.83

against 0.64 for CKA, the advantage over CKA is positive whichever encoder is left out (0.09 to 0.26), but with five encoders no valid sampling interval exists, and no index beats linear predictivity or mutual k -NN. Heads on identical features (CKA = 1) trained on different data disagree, and rescaling features at fixed weight decay shifts mean Laplace EU over three orders of magnitude while CKA stays at one. For softmax probes, EU summaries are almost a function of confidence (rank correlation -0.999 to -0.997), so EU and aleatoric agreement behave alike; this and other failed pre-registered predictions are reported in full.

1 INTRODUCTION

Neural networks trained with different objectives, architectures and data are converging toward similar representations; the platonic-representation hypothesis rests mainly on mutual nearest-neighbour overlap and its growth with model scale (Huh et al., 2024), and the broader literature uses centred kernel alignment (CKA; Kornblith et al., 2019) and canonical correlations (Raghu et al., 2017; Morcos et al., 2018). A natural but untested inference is that aligned encoders also induce heads that are unsure about the same inputs, which would make alignment a label-free tool for transferring uncertainty estimates, selecting encoders for active learning, or arguing that aligned models share blind spots. We are not aware of work that uses alignment this way explicitly; our question is whether doing so would be justified.

We study this for *epistemic* uncertainty (EU), the reducible part that reflects limited data (Kendall and Gal, 2017; Hüllermeier and Waegeman, 2021). At one level the answer is immediate: the EU of a head depends on its prior and training data, which no function of two representations can see. The useful question is

sharper: *which* information about a pair of representations controls EU agreement once the head is fixed, and do the common indices carry it? For CKA our answer is no: it discards two pieces of information that matter. It normalises away the scale of the features, and a regularised head treats scale as part of its prior. And it weights spectral directions by their squared variance, whereas EU counts every direction above the regularisation level, however small its variance.

Contributions.

1. **An exact identity** (Theorem 2): for Gaussian features and Bayesian linear heads, the correlation of two heads’ epistemic variances equals S_ρ , the cosine between ridge smoothers at the heads’ prior strengths, an index from the ridge-CCA family. CKA is its $\rho \rightarrow \infty$ limit and a normalised sum of squared canonical correlations its $\rho \rightarrow 0$ limit (Proposition 3).
2. **Why CKA fails:** CKA is neither necessary nor sufficient for EU agreement (Corollary 4); its invariances are not invariances of EU (Propositions 5 and 6); and even identical features give imperfect EU agreement once priors differ (Corollary 8). Supporting results (Proposition 10 and lemmas 7 and 9) are elementary or classical and stated for completeness.
3. **Pre-registered and additional exploratory experiments** on five frozen encoders with CIFAR-10H human labels, using bootstrap, Laplace and closed-form estimators, re-test reliabilities, leave-one-encoder-out sensitivity analyses and reliability ceilings. Confirmed and failed predictions are reported alike (Table 1).

2 SETUP AND RELATED WORK

Representations and alignment. An encoder $\phi_A : \mathcal{X} \rightarrow \mathbb{R}^{d_A}$ is frozen; inputs $x \sim P$. We centre features and write $\Sigma_A = \mathbb{E}[\phi_A \phi_A^\top]$, $\Sigma_{AB} = \mathbb{E}[\phi_A \phi_B^\top]$. Population linear CKA is $\text{CKA}(A, B) = \|\Sigma_{AB}\|_F^2 / (\|\Sigma_A\|_F \|\Sigma_B\|_F)$ (Kornblith et al., 2019); it is invariant to $\phi \mapsto cQ\phi$ for $c > 0$ and orthogonal Q . Mutual k -NN with cosine similarity (Huh et al., 2024) shares this invariance. Prior work has already shown that CKA can miss functionally relevant differences: it is insensitive to low-variance directions that matter for downstream behaviour (Ding et al., 2021) and can be manipulated without changing function (Davari et al., 2023). Our tail argument is a special case of this insensitivity, made exact for one downstream quantity. Other lines of work regularise shape metrics (Williams et al., 2021), define distances through ridge-regularised

linear prediction (GULP; Boix-Adserà et al., 2022), or interpret CKA and CCA as average alignment of optimal linear decoders (Harvey et al., 2024); see also RSA (Kriegeskorte et al., 2008). These works concern predictions or decodable information, and error consistency compares which items two models get wrong (Geirhos et al., 2020). We ask what alignment implies for the *uncertainty* of a head. Human soft labels are used here only as an evaluation target; on learning from annotator disagreement see Uma et al. (2021).

Heads and epistemic uncertainty. A head is fitted on n labelled examples with feature matrix $\Phi_A \in \mathbb{R}^{n \times d_A}$. For a Bayesian linear head with prior $w \sim \mathcal{N}(0, \tau^2 I)$ and noise variance σ^2 , the posterior variance of $w^\top \phi_A(x)$ is

$$v_A(x) = \sigma^2 \phi_A(x)^\top (\Phi_A^\top \Phi_A + \lambda I)^{-1} \phi_A(x), \quad \lambda = \sigma^2 / \tau^2, \quad (1)$$

which does not depend on the labels (Rasmussen and Williams, 2006). With $\hat{\Sigma}_A = \Phi_A^\top \Phi_A / n$ and $\rho = \lambda / n$, $nv_A(x) / \sigma^2 = \phi_A(x)^\top (\hat{\Sigma}_A + \rho I)^{-1} \phi_A(x)$. We call the population version $u_A(x) = \phi_A(x)^\top (\Sigma_A + \rho A I)^{-1} \phi_A(x)$ the *EU score*; it is the ridge leverage score of x (Alaoui and Mahoney, 2015), whose average over the training points is the effective dimension, and interpolates between a Mahalanobis leverage ($\rho \rightarrow 0$) and a squared norm ($\rho \rightarrow \infty$), connecting to Bayesian last-layer and neural-linear models (Snoek et al., 2015; Riquelme et al., 2018) and to distance-aware feature-space uncertainty (Lee et al., 2018; van Amersfoort et al., 2020; Liu et al., 2020; Mukhoti et al., 2023). In experiments we also use softmax heads with bootstrap ensembles (Efron and Tibshirani, 1993; Lakshminarayanan et al., 2017), summarised by the mutual information (Houlsby et al., 2011) and by quantile-interval widths, and last-layer Laplace (Daxberger et al., 2021). Entropy-based decompositions have known limitations (Wimmer et al., 2023), and in practice AU and EU estimators are strongly entangled (Mucsányi et al., 2024); we therefore report several EU summaries, including the closed form, which does not depend on labels.

EU agreement. For two models we measure agreement as the correlation of their EU scores across inputs; in experiments, as Spearman correlation, partial Spearman controlling for both models’ confidence and human-label entropy, and Spearman within human-entropy deciles. Partial and stratified variants reduce, but cannot remove, the trivial explanation that both models are uncertain on intrinsically ambiguous images, because human entropy is itself a noisy covariate (reliability 0.70); Section 4.2 reports a sensitivity analysis over covariate estimators.

3 THEORY

All proofs are in Appendix A. Let $w_i(\rho) = \lambda_i/(\lambda_i + \rho)$ for eigenvalues λ_i of a covariance.

Definition 1 (Spectral agreement index). For symmetric M_A, M_B let

$$S(M_A, M_B) = \frac{\text{tr}(M_A \Sigma_{AB} M_B \Sigma_{BA})}{\sqrt{\text{tr}((M_A \Sigma_A)^2) \text{tr}((M_B \Sigma_B)^2)}},$$

and $S_\rho := S((\Sigma_A + \rho A I)^{-1}, (\Sigma_B + \rho B I)^{-1})$.

On a sample, S_ρ is the Frobenius cosine between the ridge smoothers $H = \Phi(\Phi^\top \Phi + \rho I)^{-1} \Phi^\top$, computable with $d \times d$ matrices only. S_ρ is not a new similarity measure: ridge interpolation between correlation and covariance analyses is classical (Vinod, 1976; Bach and Jordan, 2002), and GULP (Boix-Adserà et al., 2022) compares representations through ridge-regularised linear predictors, which is closely related. Comparisons of optimal linear readouts also underlie CKA and CCA (Harvey et al., 2024). What we add is the exact link between this family and agreement of *posterior variances*, which identifies the head’s own prior strength as the right regularisation level.

Theorem 2 (EU agreement identity). *If $(\phi_A(x), \phi_B(x))$ is zero-mean jointly Gaussian under $x \sim P$, then for any symmetric M_A, M_B , $\text{corr}(\phi_A^\top M_A \phi_A, \phi_B^\top M_B \phi_B) = S(M_A, M_B)$. In particular $\text{corr}(u_A, u_B) = S_\rho$.*

Proposition 3 (Limits). *With $\rho_A = \rho_B = \rho$: (i) $\lim_{\rho \rightarrow \infty} S_\rho = \text{CKA}(A, B)$; (ii) if Σ_A, Σ_B are invertible, $\lim_{\rho \rightarrow 0} S_\rho = \sum_j r_j^2 / \sqrt{d_A d_B}$, where r_j are the canonical correlations of A and B ; for $d_A = d_B = d$ this is the mean squared canonical correlation (the R_{CCA}^2 similarity), and otherwise the sum of squared canonical correlations normalised by $\sqrt{d_A d_B}$.*

The proof is one application of Isserlis’ theorem; its value lies in the interpretation. Kornblith et al. (2019) already observed that CCA, linear regression and CKA differ in how they weight eigendirections; what the identity adds is the weighting that EU agreement requires. In the eigenbasis, CKA weighs a shared direction by λ_i^2 , whereas S_ρ weighs it by $w_i(\rho)^2$: every direction with $\lambda_i \gg \rho$ counts as one, however small its variance. This gives a concrete prediction: when heads are weakly regularised, whitened (CCA-type) indices should track EU agreement better than CKA, reversing the preference for CKA over CCA that Kornblith et al. (2019) established for a different target (identifying corresponding layers). Since learned representations have long, power-law spectral tails (Agrawal et al., 2022), as do population codes in visual cortex (Stringer et al., 2019), the two indices read different

parts of the spectrum. Mutual k -NN with cosine similarity sits in between: like CKA it is invariant to cQ , so it is blind to scale, but as a local, rank-based statistic it is not dominated by the top eigendirections and responds to the tail (it drops to 0.62 under the tail reshaping of Section 4.3, where CKA stays above 0.956).

Corollary 4 (CKA is neither necessary nor sufficient). *Let $\phi_A = (s, a)$, $\phi_B = (s, b)$ with $s \in \mathbb{R}^k$, $a, b \in \mathbb{R}^m$ independent Gaussians with covariances $\lambda_s I, \lambda_p I, \lambda_p I$. Then $\text{CKA} = \frac{k\lambda_s^2}{k\lambda_s^2 + m\lambda_p^2}$ and $\text{corr}(u_A, u_B) = \frac{k w_s^2}{k w_s^2 + m w_p^2}$. Hence for every $\epsilon > 0$ there are pairs with $\text{CKA} \geq 1 - \epsilon$ and $\text{corr}(u_A, u_B) \leq \epsilon$, and pairs with $\text{CKA} \leq \epsilon$ and $\text{corr}(u_A, u_B) \geq 1 - \epsilon$.*

Proposition 5 (Scale is a prior change). *For $c > 0$ and orthogonal Q , $\text{CKA}(\phi, cQ\phi) = 1$, cosine k -NN graphs are unchanged, and $v(cQ\Phi, cQx; \lambda) = v(\Phi, x; \lambda/c^2)$. Under Gaussian features, the EU agreement between the head on $cQ\phi$ and the head on ϕ at the same λ is $\sum_i w_i(\rho) w_i(\rho/c^2) / \sqrt{\sum_i w_i(\rho)^2 \sum_i w_i(\rho/c^2)^2}$, which is < 1 unless the spectrum is flat on its support.*

Proposition 6 (Tail reshaping). *Let $\Sigma = U \Lambda U^\top$, let $\mathcal{T}$ be a set of tail directions, and let T_s scale them by s . With $\eta = \sum_{i \in \mathcal{T}} \lambda_i^2 / \sum_{i \notin \mathcal{T}} \lambda_i^2$, $\text{CKA}(\phi, T_s \phi) = \frac{1+s^2\eta}{\sqrt{(1+\eta)(1+s^4\eta)}} \geq 1 - (s^2 - 1)^2 \eta$, whereas the effective dimension changes by $\sum_{i \in \mathcal{T}} [w_i(\rho/s^2) - w_i(\rho)]$.*

Lemma 7 (Mean EU is effective dimension). *On the training points, $\frac{1}{n} \sum_i v(x_i) = \frac{\sigma^2}{n} d_{\text{eff}}(\rho)$ with $d_{\text{eff}}(\rho) = \sum_i \frac{\lambda_i}{\lambda_i + \rho}$ (Zhang, 2005; Caponnetto and De Vito, 2007).*

Together, Proposition 6 and lemma 7 give a transformation under which CKA stays above $1 - (s^2 - 1)^2 \eta$ while mean EU changes by a factor of order $|\mathcal{T}|/d_{\text{eff}}$. The participation ratio $(\sum \lambda_i)^2 / \sum \lambda_i^2$, a common “effective rank”, is top-heavy like CKA and does not track $d_{\text{eff}}(\rho)$.

Corollary 8 (Identical features do not imply EU agreement). *Two heads on the same features with prior strengths $\rho_1 \neq \rho_2$ (for a summed likelihood with fixed λ , $\rho = \lambda/n$, so different sample sizes suffice) have EU agreement $\sum_i w_i(\rho_1) w_i(\rho_2) / \sqrt{\sum_i w_i(\rho_1)^2 \sum_i w_i(\rho_2)^2}$ under Gaussian features, while every alignment metric equals its maximum.*

Lemma 9 (Bootstrap is a shrunk posterior; classical). *For fixed-design ridge with residual resampling, $\text{Cov}(\hat{w}) = \sigma^2 A^{-1} G A^{-1}$ with $G = \Phi^\top \Phi$, $A = G + \lambda I$, versus the posterior $\sigma^2 A^{-1}$; along eigendirection i the ratio is $g_i/(g_i + \lambda)$.*

This is the textbook sandwich covariance of ridge regression; the gap between bootstrap and posterior

variance also motivates randomised-prior ensembles (Osband et al., 2018). Our only addition is the consequence for agreement: the bootstrap EU score is $\phi^\top(\Sigma + \rho)^{-1}\Sigma(\Sigma + \rho)^{-1}\phi$, so Theorem 2 applies with direction weights w_i^4 rather than w_i^2 . Bootstrap EU is therefore more top-heavy than posterior or Laplace EU, and we predict it moves less under tail reshaping.

Proposition 10 (In-sample resolvent bound). *For Gram matrices K, L on n observed points and noise $s^2 > 0$, the posterior covariances $P(G) = G - G(G + s^2 I)^{-1}G$ satisfy $\|P(K) - P(L)\|_{\text{op}} \leq \|K - L\|_{\text{op}}$.*

The bound uses the *unnormalised* Gram in operator norm, exactly the information that CKA discards. It does not extend with constant one to unobserved evaluation points: in random trials with partially observed points the ratio reached ≈ 5 (Appendix A). Separately, because EU combines the training design (through the posterior) with the evaluation inputs (where it is read out), we compute S_ρ on training $\cup$ evaluation inputs; a training-only variant is reported in Section 4.4.

4 EXPERIMENTS

Pre-registration. Predictions, falsifiers and gate thresholds were frozen in code before any real-data result was computed (hash in the supplement); Table 1 lists every one with its outcome. Analyses not listed there are marked exploratory.

Data, encoders and heads. We use CIFAR-10 (Krizhevsky, 2009) training images (50k, hard labels) to fit heads and the 10k test images with CIFAR-10H soft labels (about 50 annotators per image; Peterson et al., 2019) for evaluation only. Five frozen encoders differ in objective and architecture: DINOv2 ViT-B/14 (Oquab et al., 2024), CLIP ViT-B/16 image tower (Radford et al., 2021; Cherti et al., 2023), supervised ViT-B/16 (Dosovitskiy et al., 2021; Steiner et al., 2022), ConvNeXt-S (Liu et al., 2022) and MAE ViT-B/16 (He et al., 2022), all via `timm/open_clip` (Wightman, 2019). Images are upsampled from 32 to 224 pixels (a shared limitation). Features are mean-pooled patch tokens (spatial mean for ConvNeXt) of the raw block output at relative depths 0.5, 0.75, 1, centred and divided by one scalar computed on the training set; no per-dimension standardisation is applied, since it would undo the transformations of Propositions 5 and 6. Heads are softmax regressions with weight decay chosen per encoder and depth by validation NLL on a held-out 10k training split. EU is estimated with $M = 50$ Poisson-bootstrap members (quantile width $q_{.95} - q_{.05}$ summed over classes, and mutual information), with class-block last-layer Laplace, and with the closed form $u(x)$ at ρ equal to

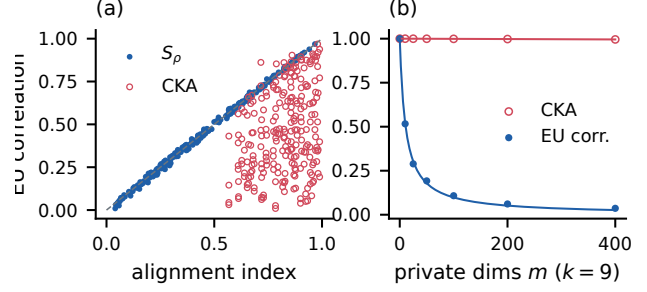


Figure 1: **(a)** Empirical EU correlation versus S_ρ (filled) and CKA (open) over 60 synthetic pairs and four ρ . **(b)** The construction of Corollary 4 ($k = 9$, $\lambda_s = 10$, $\lambda_p = 0.1$, $\rho = 0.01$): CKA stays near one as private dimensions are added while EU correlation falls as $kw_s^2/(kw_s^2 + mw_p^2)$; lines are the exact closed forms of Corollary 4.

the weight decay.

Reliability and positive control (E0). The split-half reliability ceiling of CIFAR-10H entropy is 0.70 (Spearman–Brown corrected; range over 50 random splits 0.69–0.71, which reflects split randomness, not sampling uncertainty); every AU number below should be read against it. Plug-in and Dirichlet-posterior entropy estimators agree at Spearman 0.95. In a known-AU control, where labels are drawn from a well-specified softmax teacher on each encoder’s features, model AU recovers the true AU with Spearman between 0.89 and 0.96 across encoders and temperatures, so the pre-registered gate passes.

4.1 Synthetic verification of the theory

Over 60 random representation pairs (shared and private blocks with power-law spectra, random mixings, one third with Laplace instead of Gaussian latents) and four values of ρ , the sample S_ρ predicts the empirical EU correlation with mean absolute error 0.010 (correlation 0.999), while CKA has mean absolute error 0.37 and correlation 0.32 with EU agreement (Figure 1a). The error of CKA shrinks as ρ grows (0.57 at $\rho = 0.01$ to 0.07 at $\rho = 10$), as Proposition 3 predicts. The construction of Corollary 4 with $k = 9$ shared and $m = 200$ private directions gives CKA = 0.998 with EU correlation 0.06; the converse construction gives CKA = 0.003 with EU correlation 0.96 (Figure 1b). With identical features and a fixed ridge, a head trained on 1% of the data has EU correlation 0.76 with the full-data head, below the population prediction of Corollary 8 (0.86): finite-sample covariance error adds a data term on top of the change in ρ .

Table 1: Pre-registered predictions and outcomes (frozen before any real-data result). The last column is the verdict on the *prediction*. [†]Uninformative in hindsight: the re-test partial agreement at $M = 50$ is itself ≈ 0.56 , so the threshold could not be reached. [‡]The frozen falsifier had no numeric threshold; our operationalisation is post hoc (Appendix B). For S the range is over leave-one-encoder-out subsets (Section 4.4).

id	prediction (falsifier)	observed	verdict
E0	pipeline recovers known AU (gate: Spearman ≥ 0.5)	0.89–0.96	supported
F-data	data term matters on identical features (falsifier: 0.26)		inconclusive [†]
F-AU	partial EU agr. 10% vs 100% > 0.9) [†] AU agreement protected relative to EU (falsifier: AU drop 0.10 vs 0.10 degrades as much) [‡]		not supported
E2b-scale	rescaling moves EU at fixed wd, not when re-tuned (Prop. 5)	Laplace $\times 0.02$ –62	supported
E2b-tail	bootstrap EU moves less than Laplace EU 6/6 (Lemma 9)		supported
E4	head and data dominate at high alignment (falsifier: encoder share $> 80\%$)	93%	not supported
S	S_ρ predicts EU agreement better than CKA (falsifier: no better) [†]	0.83 vs 0.64; LOEO diff. $+0.09$ to $+0.26$	inconclusive
Spec	$d_{\text{eff}}(\rho)$ predicts mean EU	closed form 0.90; bootstrap -0.19	partly supported

Table 2: E2: heads on *identical* frozen features (all alignment metrics = 1). Item-level Spearman agreement, mean (min–max) over five encoders; partial = controlling for both models’ confidence and human entropy.

pair	EU width	EU partial	AU
re-test floor	1.00 (0.99–1.00)	0.56	1.00 (1.00–1.00)
25% vs 100% data	0.94 (0.90–0.96)	0.36	0.94 (0.91–0.97)
10% vs 100% data	0.90 (0.84–0.94)	0.26	0.90 (0.85–0.94)
disjoint halves	0.91 (0.88–0.94)	0.36	0.92 (0.89–0.95)
wd $\div 10$	0.98 (0.98–0.99)	0.24	0.98 (0.97–0.99)
wd $\times 10$	0.97 (0.97–0.98)	0.49	0.97 (0.96–0.98)

4.2 Identical features, different heads (E2)

Within each encoder the paired heads see literally the same features, so CKA and every other alignment index equals one. Raw EU agreement is high for all pairs (Table 2), but this is shared confidence: for softmax probes the bootstrap width has rank correlation -0.999 to -0.997 with the max-probability and at least 0.994 with AU on every encoder, whereas the closed-form $u(x)$, which ignores labels, correlates with it at only -0.08 to 0.31. We therefore read E2 through the partial agreement, controlling for both heads’ confidence and human entropy. It falls from a re-test value of 0.56 (two ensembles with identical configuration) to 0.36 for disjoint halves of the training set and 0.26 for 10% versus 100% of the data (Figure 3 in the supplement). Because the head objective averages the loss, the weight decay sets the same ρ at every sample size, so this drop reflects the finite-sample data term (Section 4.1) rather than the change of ρ in Corollary 8; weight decay was not re-selected for subset heads. Falsifier F-data (> 0.9) is not triggered, but in hindsight

it could not have fired, since the re-test value itself is far below 0.9 (Table 1).

Estimator noise and redraws (exploratory).

The re-test partial agreement is limited by Monte Carlo noise: for DINOv2 it rises from 0.45 at $M = 10$ to 0.60 at $M = 50$ and 0.82 at $M = 200$ (Figure 2a). We therefore repeated the identical-features analysis at $M = 200$ for 5 encoders, with three independent subset draws and the two weight-decay pairs, and divided each partial agreement by the geometric mean of the two heads’ own re-test reliabilities (Table 3). The reliability-corrected EU agreement is 0.43 for 10% versus 100% of the data and 0.63 for disjoint halves; across encoders and draws the 95% item-bootstrap intervals reach at most 0.58 and 0.79, so the disagreement is not estimator noise. Two caveats bound the interpretation. First, after the controls the residual holds only 0.2%–0.6% of the rank variance of the bootstrap width: the finding concerns the fine structure of EU beyond confidence, not which inputs are most uncertain. Second, this residual has no external validation as an epistemic quantity. At the decision level, the overlap of the 10% highest-uncertainty (acquisition) sets is 0.93 between re-test ensembles but 0.70 between the 10% and 100% heads ($M = 50$). Replacing the plug-in human entropy by the Dirichlet-posterior estimator or by two independent split-half entropies changes the partial agreements by at most 0.001, because confidence absorbs almost all shared variance.

Falsifier F-AU is **triggered**: AU agreement on identical features degrades about as much as EU agreement (raw drop 0.10 vs. 0.10; corrected agreement at $M = 200$: AU 0.48 vs. EU 0.43). The pre-registered contrast that AU agreement is protected is not supported for

Table 3: Heads on identical features at $M = 200$ (exploratory): mean over 5 encoders (and 3 subset draws where applicable). Partial = controlling for both heads’ confidence and human entropy; corrected = partial divided by the geometric mean of the two heads’ re-test partial reliabilities, with the range of 95% item-bootstrap intervals across encoders and draws; resid. = fraction of rank variance of EU width left after the controls.

pair		raw	partial	corrected	[95% range]	resid.
re-test	EU width	1.00	0.79	1		0.3%
	AU	1.00	0.94	1		0.2%
wd $\div 10$	EU width	0.99	0.35	0.42	[0.15, 0.87]	0.3%
	AU	0.99	0.29	0.31	[-0.05, 0.90]	0.2%
wd $\times 10$	EU width	0.98	0.63	0.74	[0.53, 0.84]	0.3%
	AU	0.98	0.57	0.59	[0.23, 0.81]	0.2%
disjoint halves	EU width	0.92	0.47	0.63	[0.44, 0.79]	0.3%
	AU	0.92	0.53	0.57	[0.42, 0.70]	0.3%
10% vs 100%	EU width	0.90	0.34	0.43	[0.20, 0.58]	0.3%
	AU	0.91	0.45	0.48	[0.30, 0.65]	0.2%

softmax probes, consistent with the near-collinearity of their AU and EU summaries. In an exploratory observation, AU’s relation to the human target is stable: per encoder, Spearman between model AU and human entropy varies by at most 0.05 across the eight head configurations, at 0.34–0.43 overall (0.40–0.51 of the attainable maximum $\sqrt{0.70} = 0.84$).

4.3 Transformations that preserve alignment (E2b)

Applying $\phi \mapsto cQ\phi$ leaves CKA and mutual k -NN at one for both encoders tested (DINOv2 and CLIP; Table 4). At fixed weight decay, EU moves with c : across $c \in [0.1, 10]$ the mean Laplace logit variance changes by a factor between 0.017 and 61.7, the mean bootstrap width between 0.38 and 1.84, and EU rankings change (Laplace MI rank agreement down to 0.72, bootstrap width down to 0.89). When the weight decay is re-tuned by validation, the selected value scales as c^2 (from 10^{-5} at $c = 0.1$ to 10^{-1} at $c = 10$), which is exactly the prior change of Proposition 5, and the largest deviation of any EU rank agreement or mean ratio from the untransformed head is 0.086 (the re-tuned grid is coarse, one decade per step). The effect therefore operates through the prior, as Proposition 5 says; it vanishes when the weight decay is re-tuned. The point is not that practitioners will meet this failure after tuning, but that CKA cannot tell whether two heads’ priors are matched to their feature scales. Tail reshaping (scaling the directions outside the top 95% of variance by $s \in \{0.5, 2, 4\}$) keeps CKA at or above 0.956 while the mean Laplace logit variance changes by up to a factor 2.3 and the mean bootstrap width by up to 1.39 (ranges over both encoders); mu-

tual k -NN, which is sensitive to the tail, drops to 0.62. The closed-form EU barely moves (mean ratio within 0.04 of one): at the selected weight decay the heads are in the small- ρ regime, where $u(x)$ approaches the leverage, which is invariant to every invertible linear map. The softmax estimators move because their effective prior strength is set by the curvature of the likelihood, not by the weight decay alone. Lemma 9 predicts that bootstrap EU moves less than Laplace EU under tail reshaping; this held in 6 of 6 settings (3 values of $s \times 2$ encoders, comparing rank changes). The lemma is derived for residual resampling, so this is a heuristic extrapolation to Poisson-resampled softmax heads.

4.4 Across encoders: which index tracks EU agreement?

For all 10 encoder pairs at three matched relative depths (30 pairs) we compute CKA, mutual k -NN ($k = 10$), symmetric linear predictivity, S_ρ at the two heads’ own ρ and its $\rho \rightarrow 0$ limit, on training $\cup$ test inputs, together with item-level EU agreement (Table 5; scatter plots in Figure 4 of the supplement). For partial bootstrap EU agreement, S_ρ , $S_{\rho \rightarrow 0}$ and linear predictivity all reach 0.83, mutual k -NN 0.72 and CKA 0.64. This ordering is what Proposition 3 predicts for weakly regularised heads, but the evidence is limited: pairs share encoders, and with five encoders no resampling scheme gives a valid interval. We therefore report how the comparison changes when each encoder is left out (Table 5): $S_\rho - \text{CKA}$ stays positive in 5/5 subsets (0.09 to 0.26) for partial agreement and in 4/5 for closed-form agreement, whereas the differences from linear predictivity (-0.08 to +0.05) and mutual k -NN (-0.04 to +0.17) change sign. S_ρ is also indistinguishable from its own $\rho \rightarrow 0$ limit, so at the selected weight decays we cannot attribute any advantage to the head’s prior strength. Falsifier S (“no better than CKA”) is not triggered on the point estimate; we classify the prediction as inconclusive. The three indices that do well are all tail-sensitive, and two of them (CCA-type, mutual k -NN) are as scale-blind as CKA: the comparison isolates spectral weighting, not scale. CKA is not uninformative, and for raw agreement linear predictivity (0.41) is weaker still; the highest-CKA pair at the final depth (CKA 0.837) reaches a partial EU agreement of only 0.21. S_ρ , its $\rho \rightarrow 0$ limit and mutual k -NN predict AU agreement about as well as EU agreement (S_ρ : 0.92; linear predictivity is the exception), so they index shared head behaviour rather than EU specifically. On real features Theorem 2 holds only ordinally: S_ρ ranks the *Pearson* agreement of the closed form, the quantity in Theorem 2 (Table 5 uses Spearman), at 0.93 (CKA: 0.74) but its level is off by 0.34. The offset is due to non-Gaussianity: replacing each pair’s features by Gaussian draws with the same joint

Table 4: E2b (fixed weight decay): alignment of the transformed to the original features, Spearman rank agreement of EU with the untransformed head, and mean EU relative to it, for both encoders (width = bootstrap quantile width; Laplace = MI for the rank, logit variance for the ratio). The text summarises ranges over both encoders.

transform	DINOv2-B						CLIP-B/16					
	CKA	mKNN	rank width	agr. Lapl.	mean width	ratio Lapl.	CKA	mKNN	rank width	agr. Lapl.	mean width	ratio Lapl.
$c=0.1$	1.000	1.00	0.94	0.89	0.56	0.017	1.000	1.00	0.89	0.85	0.38	0.0301
$c=0.3$	1.000	1.00	0.98	0.94	0.74	0.125	1.000	1.00	0.96	0.95	0.62	0.182
$c=1.0$	1.000	1.00	1.00	1.00	1.00	1	1.000	1.00	1.00	1.00	1.00	1
$c=3.0$	1.000	1.00	0.99	0.91	1.29	6.78	1.000	1.00	0.98	0.92	1.45	4.24
$c=10.0$	1.000	1.00	0.97	0.85	1.52	61.7	1.000	1.00	0.95	0.72	1.84	39.3
tail $s=0.5$	1.000	0.96	1.00	1.00	1.02	0.894	1.000	0.95	1.00	0.99	0.98	0.788
tail $s=2.0$	0.998	0.89	1.00	0.99	1.07	1.27	0.999	0.86	0.99	0.98	1.13	1.47
tail $s=4.0$	0.956	0.68	0.98	0.95	1.30	1.8	0.982	0.62	0.97	0.92	1.39	2.28

Table 5: Across 30 encoder pairs (5 encoders $\times$ 3 depths): Spearman correlation between each alignment index and item-level agreement, on all pairs and (in parentheses) its range over the five leave-one-encoder-out subsets of 18 pairs. With five encoders no valid sampling interval is available; the ranges are descriptive. Bottom rows: differences, with the number of subsets in which they are positive.

index	width (partial)	closed form	AU
CKA	0.64 (0.62–0.67)	0.76 (0.62–0.89)	0.68 (0.54–0.84)
mutual k -NN	0.72 (0.69–0.89)	0.90 (0.85–0.95)	0.86 (0.75–0.91)
linear predictivity	0.83 (0.79–0.90)	0.56 (0.35–0.74)	0.53 (0.39–0.78)
$S_{\rho \rightarrow 0}$	0.83 (0.77–0.90)	0.93 (0.87–0.96)	0.93 (0.89–0.92)
S_{ρ} (head’s ρ)	0.83 (0.76–0.90)	0.92 (0.84–0.95)	0.92 (0.87–0.92)
S_{ρ} –CKA	+0.19 (+0.09 to +0.26; 5/5 > 0)	+0.16 (-0.02 to +0.30; 4/5 > 0)	+0.24 (+0.08 to +0.33; 5/5 > 0)
S_{ρ} –mutual k -NN	+0.11 (-0.04 to +0.17; 4/5 > 0)	+0.03 (-0.08 to +0.09; 3/5 > 0)	+0.07 (+0.00 to +0.12; 4/5 > 0)
S_{ρ} –linear pred.	-0.00 (-0.08 to +0.05; 1/5 > 0)	+0.37 (+0.11 to +0.52; 5/5 > 0)	+0.40 (+0.13 to +0.53; 5/5 > 0)

covariance reduces the mean absolute error between S_{ρ} and the closed-form Pearson agreement from 0.34 to 0.019, so the identity is exact up to higher cumulants.

Does the head’s ρ matter? (exploratory). At the selected weight decays all heads have $d_{\text{eff}}(\rho)/d > 0.98$, so $S_{\rho} \approx S_{\rho \rightarrow 0}$. We therefore refit all 15 encoder–depth heads ($M = 20$) at a common weight decay $\rho \in \{10^{-3}, 10^{-2}, 10^{-1}, 1\}$, which lowers $d_{\text{eff}}(\rho)/d$ to 0.06–0.29 at $\rho = 1$ (Figure 2b). As predicted, the advantage of evaluating the index at the matched ρ grows with ρ : for closed-form EU agreement $S_{\rho} - S_{\rho \rightarrow 0}$ moves from -0.02 at $\rho = 10^{-3}$ to +0.05 at $\rho = 1$ (0.94 vs. 0.89; CKA 0.69), and for partial bootstrap agreement to +0.08 (0.96 vs. 0.88; CKA 0.71). At $\rho = 1$ the difference is positive in 5/5 (closed form) and 5/5 (partial bootstrap) leave-one-encoder-out subsets, but strongly regularised heads underfit for some encoders (accuracy 0.50–0.97), and the weight decay of a softmax head is not its effective prior strength (its curvature-calibrated prior strength $\rho_{\text{eff}} = \text{wd}/\bar{h}$, with $\bar{h}$ the mean $p(1-p)$ of the MAP head, is a median 120 times larger), so this is a trend, not an established effect. Computing S_{ρ} on training inputs only instead of training $\cup$

test changes its correlation with EU agreement by at most 0.03. Evaluating S_{ρ} at the curvature-calibrated ρ_{eff} of each softmax head (median 0.12, range 0.003–0.52) does not help either: it correlates with partial bootstrap agreement at 0.78, below its $\rho \rightarrow 0$ limit (0.83), so for softmax probes we cannot confirm the matched-prior prediction. Between different encoders, the top-10% acquisition sets overlap by only 0.38 on average.

Label-free prediction. Across encoders and depths, $d_{\text{eff}}(\rho)$ from unlabelled training features rank-correlates with mean closed-form EU at 0.90, as Lemma 7 predicts, but not with the mean bootstrap width of softmax heads (-0.19), whose magnitude follows accuracy; this pre-registered prediction failed for softmax heads.

Swap design (E4). In a $2 \times 2 \times 2$ design on the highest-CKA pair (DINOv2-B vs. supervised ViT-B/16, CKA 0.837; encoder $\times$ weight decay $\{w, 10w\} \times$ disjoint training half), the encoder accounts for 93% of the Shapley decomposition of raw EU disagreement, the training half for 9% and the weight decay for essen-

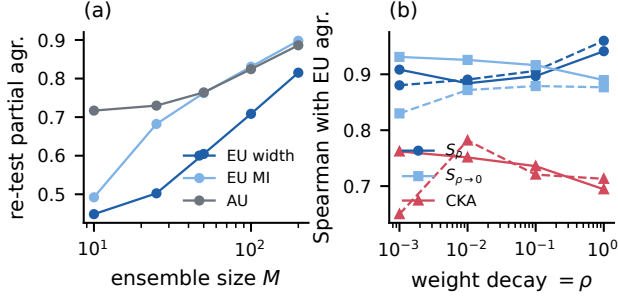


Figure 2: Exploratory analyses. (a) Re-test partial agreement of two independent ensembles versus ensemble size M (DINOv2, final layer; seeds independent of Table 2, hence 0.60 vs. 0.61 at $M = 50$): the noise floor in Table 2 is a Monte Carlo limit. (b) Weight-decay sweep over 30 encoder pairs: Spearman correlation of S_ρ at the matched ρ , its $\rho \rightarrow 0$ limit and CKA with closed-form (solid) and partial bootstrap (dashed) EU agreement.

tially none (-2%). This triggers the pre-registered E4 falsifier ($> 80\%$): among the encoders we could test, alignment never reached a regime where head and data terms dominate. In an exploratory variant using partial disagreement, the shares are 61%, 30% and 10% for encoder, data and weight decay (re-test disagreement 0.33).

5 DISCUSSION

What the results say. That a representation-only index cannot determine EU follows from Equation (1); our contribution is to say which information an index needs and why CKA lacks it. Theorem 2 identifies the relevant summary for linear-Gaussian heads, spectral overlap above the head’s prior strength weighted by $w_i(\rho)^2$, and is compatible with two observations: tail-sensitive indices (CCA-type, mutual k -NN, linear predictivity) tracked EU agreement better than CKA for weakly regularised heads, and the matched- ρ index gained ground as regularisation increased. Because every well-performing index except S_ρ is scale-invariant, the cross-encoder data support the spectral-weighting mechanism, not the scale mechanism; the latter is supported only by the controlled transformations of Section 4.3. With five encoders neither cross-encoder difference is statistically established. Two further observations are diagnostic rather than practical. CKA’s invariance to scale hides a prior-scale mismatch that moves EU by orders of magnitude (pre-registered, E2b), although the mismatch disappears once the weight decay is tuned, as Proposition 5 implies. And in exploratory redraws, identical features leave the confidence-residual EU ranking dependent on

the training data (Section 4.2); this residual is a small part of the total rank variance, so the observation concerns fine structure rather than which inputs are most uncertain. Between different encoders, however, the representation is the dominant source of EU disagreement (E4), so our results do not say that the encoder is irrelevant; they say that CKA does not tell us how EU will agree. For softmax probes, EU and AU summaries are nearly collinear with confidence, so the empirical part of this paper cannot separate EU-specific from general head agreement; the EU-specific mechanisms are established by the theory and the closed form.

Recommendations. (i) *Supported:* do not use CKA to transfer or validate EU. *Conjecture, not established here:* tail-sensitive indices are better proxies; among them, only S_ρ evaluated at the head’s prior strength on the head’s own feature scaling also reflects a prior-scale mismatch ($S_\rho(cQ\phi) = S_{\rho/c^2}(\phi)$). Whitened indices can be unstable when the dimension is large relative to the sample, which S_ρ regularises. We found no advantage of S_ρ over linear predictivity or mutual k -NN at the weight decays practitioners select. (ii) Report the weight decay together with the feature normalisation used (here a single scalar on centred features), since the pair determines the effective prior; for Laplace heads, report the prior precision relative to the trace of the feature covariance. (iii) Report confidence-controlled (partial) agreement with re-test reliabilities, since raw EU agreement of softmax probes mostly reflects shared confidence and the partial measure is noisy at common ensemble sizes.

Limitations. The theory is exact for Gaussian features and linear-Gaussian heads; on real features only its ordinal predictions held. Softmax heads enter through bootstrap and Laplace approximations; the bootstrap lemma is for residual resampling with fixed design, and our Laplace drops cross-class curvature. Five encoders give 30 dependent pairs, which limits every cross-encoder conclusion, and the encoder set has no model-scale axis, so we cannot speak to the scale-dependent convergence trend of Huh et al. (2024). For softmax probes our empirical EU summaries are confounded with confidence; an EU target that separates from confidence (e.g. out-of-distribution inputs or held-out classes) and closed-form or Gaussian-process heads as the primary empirical object are the natural next steps. We study frozen encoders and linear probes on one dataset with human labels (CIFAR-10H) at low native resolution and only in-distribution; out-of-distribution inputs, where shared blind spots matter most, fine-tuned models, other modalities and the DCIC datasets are not covered, and we do not evaluate downstream active-learning per-

formance. The positive control replaces the planned DCIC synthetic dataset with a known softmax teacher on each encoder’s features. Additional analyses not in the pre-registration (Figure 2, Table 3, leave-one-encoder-out ranges) are exploratory. Planned analyses that were not run (class/residual decomposition, AU-versus-function agreement, robustness to k , pooling and probe class) are not claimed.

AI use statement

We used generative AI tools (a large language model coding and writing assistant) for implementation of the experiment code from the authors’ design scaffold, numerical verification of theoretical claims, drafting of proofs, drafting and editing of the manuscript text, generation of figures and tables, assembly of the bibliography, and a simulated multi-reviewer critique whose findings informed this revision. The research question, pre-registered predictions and falsifiers, and experimental design were specified by the authors before AI assistance. All AI-generated code was checked against independent reference implementations by unit tests (17 tests, including 9 theory checks), every proof was checked numerically on random instances, and every number in the paper is generated by script from logged results. We take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

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CHECKLIST

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes] Sections 2 and 3.
 - (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes] S_ρ needs only $d \times d$ matrices, $O(nd^2 + d^3)$ (Section 3); compute in Appendix B.
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes] Supplementary code with `requirements.txt`.
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. [Yes] Each statement lists its assumptions (Gaussian features, linear-Gaussian heads, fixed design where relevant).
 - (b) Complete proofs of all theoretical results. [Yes] Appendix A.
 - (c) Clear explanations of any assumptions. [Yes] Sections 3 and 5.
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes] Supplementary code; public datasets and weights.
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes] Section 4 and appendix B.
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes] Appendix B; re-test floors quantify bootstrap seed variability.
 - (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes] Appendix B.
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
 - (a) Citations of the creator If your work uses existing assets. [Yes]
 - (b) The license information of the assets, if applicable. [Yes] Appendix B.
 - (c) New assets either in the supplemental material or as a URL, if applicable. [Yes] Code only.
 - (d) Information about consent from data providers/curators. [Not Applicable]
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable]
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
 - (a) The full text of instructions given to participants and screenshots. [Not Applicable] We use the public CIFAR-10H labels.
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable]
 - (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable]

When Do Aligned Representations Agree on What They Do Not Know?

Supplementary Materials

A PROOFS

Throughout, $w_i(\rho) = \lambda_i/(\lambda_i + \rho)$, and M matrices are symmetric.

Proof of Theorem 2. Write $a = \phi_A(x)$, $b = \phi_B(x)$, $P = M_A$, $Q = M_B$. For a zero-mean Gaussian vector, Isserlis' theorem (Isserlis, 1918) gives $\mathbb{E}[a_i a_j b_k b_l] = (\Sigma_A)_{ij}(\Sigma_B)_{kl} + (\Sigma_{AB})_{ik}(\Sigma_{AB})_{jl} + (\Sigma_{AB})_{il}(\Sigma_{AB})_{jk}$. Contracting with $P_{ij}Q_{kl}$ and using the symmetry of P, Q ,

$$\mathbb{E}[a^\top P a b^\top Q b] = \text{tr}(P\Sigma_A) \text{tr}(Q\Sigma_B) + 2 \text{tr}(P\Sigma_{AB}Q\Sigma_{BA}).$$

Since $\mathbb{E}[a^\top P a] = \text{tr}(P\Sigma_A)$, $\text{Cov}(a^\top P a, b^\top Q b) = 2 \text{tr}(P\Sigma_{AB}Q\Sigma_{BA})$. Setting $b = a$, $Q = P$ gives $\text{Var}(a^\top P a) = 2 \text{tr}((P\Sigma_A)^2)$, and likewise for b . Dividing gives $S(M_A, M_B)$. For u_A, u_B take $M_A = (\Sigma_A + \rho_A I)^{-1}$, $M_B = (\Sigma_B + \rho_B I)^{-1}$. $\square$

Proof of Proposition 3. (i) As $\rho \rightarrow \infty$, $\rho M_A \rightarrow I$ and $\rho M_B \rightarrow I$. Multiplying numerator and both factors of the denominator by ρ^2 , the numerator tends to $\text{tr}(\Sigma_{AB}\Sigma_{BA}) = \|\Sigma_{AB}\|_F^2$ and the denominator to $\sqrt{\text{tr}(\Sigma_A^2) \text{tr}(\Sigma_B^2)} = \|\Sigma_A\|_F \|\Sigma_B\|_F$. (ii) As $\rho \rightarrow 0$, $M_A \rightarrow \Sigma_A^{-1}$; the numerator tends to $\text{tr}(\Sigma_A^{-1} \Sigma_{AB} \Sigma_B^{-1} \Sigma_{BA}) = \|\Sigma_A^{-1/2} \Sigma_{AB} \Sigma_B^{-1/2}\|_F^2 = \sum_j r_j^2$, and $\text{tr}((\Sigma_A^{-1} \Sigma_A)^2) = d_A$. The limit is therefore $\sum_j r_j^2 / \sqrt{d_A d_B}$, which equals the mean squared canonical correlation only when $d_A = d_B$. $\square$

Proof of Corollary 4. $\Sigma_{AB} = \text{diag}(\lambda_s I_k, 0)$ and $M_A = \text{diag}((\lambda_s + \rho)^{-1} I_k, (\lambda_p + \rho)^{-1} I_m)$, so the numerator of S_ρ is $k w_s^2$ and $\text{tr}((M_A \Sigma_A)^2) = k w_s^2 + m w_p^2$; CKA follows the same way with λ in place of w . *High CKA, low agreement:* take $\rho \leq \lambda_p$ (so $w_p \geq \frac{1}{2}$, $w_s \leq 1$) and $m \geq 4k/\epsilon$; then $\text{corr} \leq k/(k + m/4) \leq \epsilon$. Then take $\lambda_p/\lambda_s \leq \sqrt{\epsilon k/m}$, so $\text{CKA} = 1/(1 + (m/k)(\lambda_p/\lambda_s)^2) \geq 1/(1 + \epsilon) \geq 1 - \epsilon$. *Low CKA, high agreement:* take $\rho \leq \lambda_s$ (so $w_s \geq \frac{1}{2}$) and $k \geq 4m/\epsilon$; then $\text{corr} = 1/(1 + m w_p^2/(k w_s^2)) \geq 1 - 4m/k \geq 1 - \epsilon$. Then take $\lambda_p/\lambda_s \geq \sqrt{k/(m\epsilon)}$, so $\text{CKA} \leq k \lambda_s^2/(m \lambda_p^2) \leq \epsilon$. $\square$

Proof of Proposition 5. CKA and cosine similarities are invariant to $\phi \mapsto cQ\phi$. For the variance, $v(cQ\Phi, cQx; \lambda) = \sigma^2 c^2 x^\top Q^\top (c^2 Q\Phi^\top \Phi Q^\top + \lambda I)^{-1} Qx = \sigma^2 x^\top (\Phi^\top \Phi + (\lambda/c^2)I)^{-1} x$. So the transformed head is the original head with prior strength ρ/c^2 . Apply Theorem 2 to the same features with $M_A = (\Sigma + \rho c^{-2} I)^{-1}$, $M_B = (\Sigma + \rho I)^{-1}$, $\Sigma_{AB} = \Sigma$: the numerator is $\sum_i w_i(\rho/c^2) w_i(\rho)$ and the denominators are $\sum_i w_i(\cdot)^2$. By Cauchy-Schwarz the ratio equals one iff $w_i(\rho)/w_i(\rho/c^2) = (\lambda_i + \rho/c^2)/(\lambda_i + \rho)$ is constant over the support, i.e. iff the nonzero λ_i are equal (for $c \neq 1$). $\square$

Proof of Proposition 6. With $T_s = U \text{diag}(s_i) U^\top$ ($s_i = s$ on $\mathcal{T}$, 1 otherwise), $\mathbb{E}[\phi(T_s \phi)^\top] = U \Lambda \text{diag}(s_i) U^\top$ and $\text{Cov}(T_s \phi) = U \Lambda \text{diag}(s_i^2) U^\top$. Hence $\text{CKA} = \sum \lambda_i^2 s_i^2 / \sqrt{\sum \lambda_i^2 \sum \lambda_i^2 s_i^4} = (1 + s^2 \eta) / \sqrt{(1 + \eta)(1 + s^4 \eta)}$. Moreover $1 - \text{CKA}^2 = \eta(s^2 - 1)^2 / ((1 + \eta)(1 + s^4 \eta)) \leq \eta(s^2 - 1)^2$, and $\text{CKA} \geq \text{CKA}^2$ since $\text{CKA} \leq 1$. The transformed eigenvalues are $s^2 \lambda_i$ on $\mathcal{T}$, and $s^2 \lambda_i / (s^2 \lambda_i + \rho) = w_i(\rho/s^2)$. $\square$

Proof of Lemma 7. $\sum_i v(x_i) = \sigma^2 \text{tr}(\Phi(\Phi^\top \Phi + \lambda I)^{-1} \Phi^\top) = \sigma^2 \sum_i g_i / (g_i + \lambda)$ with $g_i = n \hat{\lambda}_i$ the eigenvalues of $\Phi^\top \Phi$; divide by n and use $\lambda/n = \rho$. $\square$

Proof of Corollary 8. Theorem 2 with $\Sigma_A = \Sigma_B = \Sigma_{AB} = \Sigma$, $M_j = (\Sigma + \rho_j I)^{-1}$. $\square$

Proof of Lemma 9. With $y = \Phi w + \varepsilon$, $\text{Cov}(\varepsilon) = \sigma^2 I$ and $\hat{w} = A^{-1} \Phi^\top y$, $\text{Cov}_\varepsilon(\hat{w}) = \sigma^2 A^{-1} \Phi^\top \Phi A^{-1} = \sigma^2 A^{-1} G A^{-1}$. In the eigenbasis of G this is $\sigma^2 g_i / (g_i + \lambda)^2$, against the posterior $\sigma^2 / (g_i + \lambda)$. Residual resampling estimates this noise covariance as $n \rightarrow \infty$. The EU score is $\phi^\top \text{Cov}(\hat{w}) \phi = \frac{\sigma^2}{n} \phi^\top (\hat{\Sigma} + \rho)^{-1} \hat{\Sigma} (\hat{\Sigma} + \rho)^{-1} \phi$, so in Theorem 2 $M\Sigma$ has eigenvalues w_i^2 and the direction weights are w_i^4 . $\square$

Proof of Proposition 10. G and $(G + s^2 I)^{-1}$ commute, so $P(G) = G(G + s^2 I)^{-1}[(G + s^2 I) - G] = s^2(I - s^2(G + s^2 I)^{-1})$. Hence $P(K) - P(L) = s^4[(L + s^2 I)^{-1} - (K + s^2 I)^{-1}] = s^4(K + s^2 I)^{-1}(K - L)(L + s^2 I)^{-1}$, and $\|(G + s^2 I)^{-1}\|_{\text{op}} \leq s^{-2}$ for $G \succeq 0$. $\square$

The bound fails for unobserved points. For $Z = \mathcal{T} \cup \mathcal{E}$ with only $\mathcal{T}$ observed, $P(G) = G - G_{Z\mathcal{T}}(G_{\mathcal{T}\mathcal{T}} + s^2 I)^{-1} G_{\mathcal{T}Z}$. In 3000 random trials (feature dimension 1–11, $|\mathcal{T}| \in [3, 30]$, $|\mathcal{E}| \in [1, 20]$, $s^2 \in [10^{-2}, 10]$) the ratio $\|P(K) - P(L)\|_{\text{op}} / \|K - L\|_{\text{op}}$ reached 4.97, so no constant-one bound holds there.

B EXPERIMENTAL DETAILS

Pre-registration. The PREREG dictionary (claims, falsifiers, gate) was frozen before any real-data computation; its SHA-256 is 05aa7d680e7d7f11ec6ccca283c8c56ddf021a971ae97de23d94528935d3cc70 (file PREREG_FROZEN.txt in the supplementary code, frozen 2026-10-02 before any real-data computation; no external registry was used, so the hash certifies content, not time). The frozen falsifiers for E2-b (“AU degrades as much as EU”) and S (“no better than CKA”) carried no numeric threshold; the operational rules in Table 1 were fixed by us when computing them and are therefore post hoc. Analyses not in the pre-registration (the S_ρ evaluation at $\rho \rightarrow 0$, the d_{eff} -versus-participation-ratio contrast, the exploratory E4 variant, and all post-review analyses in Appendix C) are exploratory.

Features. Taps at relative depths 0.5, 0.75, 1 of the flattened block list; mean over patch tokens excluding prefix tokens (ViTs) or spatial mean (ConvNeXt) of the raw block output; fp16 cache; bicubic upsampling $32 \rightarrow 224$ with each encoder’s own normalisation. DINOv2 uses `img_size=224` with interpolated position embeddings.

Heads. Softmax regression, objective $\sum_i w_i \text{CE}_i / \sum_i w_i + \frac{\text{wd}}{2} \|W\|^2$, full-batch L-BFGS (strong Wolfe, ≤ 200 iterations), all M members trained in one tensor; agreement with scikit-learn’s optimum to 5×10^{-4} in probability. Weight decay grid $\{10^{-5}, \dots, 10^{-1}\}$, chosen by validation NLL on a fixed 10k split of the training set with hard labels. Poisson(1) bootstrap weights. EU summaries: summed 5–95% quantile width over classes and mutual information; AU: mean member entropy. Laplace: class-block-diagonal GGN at the MAP of member 0 with prior precision $n \cdot \text{wd}$, 100 Monte Carlo logit samples. Closed-form EU: $u(x) = \phi^\top (\hat{\Sigma} + \rho I)^{-1} \phi$ with $\rho = \text{wd}$.

Ensemble size. M-stability (DINOv2-B, final layer): Spearman between independent ensembles of size M and $2M$: $M = 10$: 0.992, $M = 25$: 0.997, $M = 50$: 0.998, $M = 100$: 0.999. We use $M = 50$.

Statistics. Item-level Spearman; partial Spearman on ranks residualised on both models’ confidence (max mean probability) and human plug-in entropy; stratified Spearman within human-entropy deciles. CIFAR-10H ceiling: split votes into halves without replacement (multivariate hypergeometric), Spearman between half entropies, Spearman–Brown corrected, 50 repetitions.

Compute. One Apple M2 Pro laptop (16 GB), PyTorch MPS for feature extraction (68–113 images/s) and CPU float32/float64 for heads and linear algebra.

Licences. CIFAR-10H: CC BY-NC-SA 4.0. Model weights via `timm` and `open_clip` under their respective licences.

C POST-REVIEW ANALYSES

All analyses in this section were added after an internal round of review and are exploratory.

Leave-one-encoder-out sensitivity. The 30 encoder pairs share encoders, so they are not independent, and with five encoders a cluster bootstrap is degenerate (few distinct resamples, many with fewer than three pairs). We therefore recompute every correlation on the five subsets that leave out one encoder (18 pairs each) and report the range and the number of subsets in which a difference is positive. These ranges describe sensitivity; they are not confidence intervals. An earlier draft reported percentile cluster-bootstrap intervals, which we withdrew for this reason.

Reliability and redraws. For each encoder (final depth, selected weight decay, $M = 50$) we draw three independent 10% subsets and three independent splits into halves; every head is fitted twice with different Poisson weights. A pair’s corrected agreement is its partial agreement divided by $\sqrt{r_{xx}r_{yy}}$, where r_{xx} is the partial agreement between the two replicates of head x . The re-test ensembles use seeds 0 and 1 of the full-data head. In the E2b re-tuning, the selected weight decays lie on the boundaries of the searched grid for $c = 0.1$ and $c = 10$, so the re-tuned values are bounds rather than interior optima.

Weight-decay sweep. All 15 encoder–depth heads are refitted with $M = 20$ at a common weight decay $\rho \in \{10^{-3}, 10^{-2}, 10^{-1}, 1\}$ (no validation selection), and agreement and indices are recomputed for the 30 pairs. Training-only S_ρ uses the same 10k training subset without test inputs.

D ADDITIONAL RESULTS

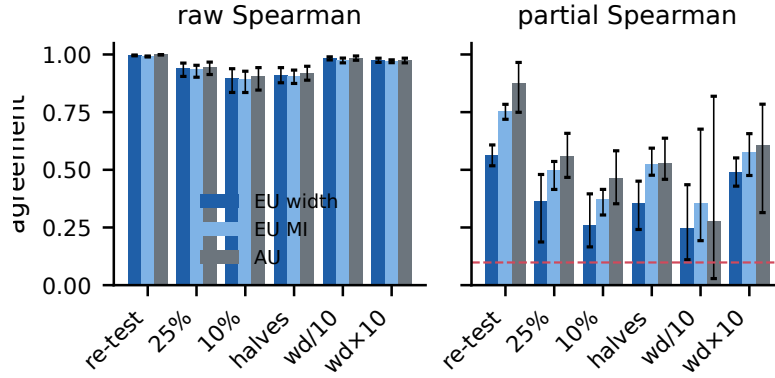


Figure 3: E2: agreement between heads on identical features, mean over five encoders (bars: min–max). Right: partial Spearman controlling for both heads’ confidence and human entropy; the dashed line is the mean partial EU agreement between *different* encoders at the final depth.

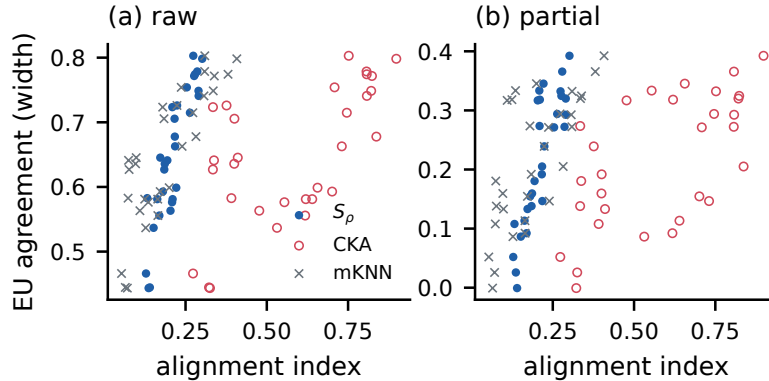


Figure 4: Across 30 encoder pairs (5 encoders, 3 depths): item-level bootstrap EU agreement, raw (a) and partial (b), against S_ρ at the heads’ ρ (filled), CKA (open) and mutual k -NN (crosses).

Table 6: E2 per encoder: Spearman (partial) agreement for EU width, mutual information and AU, and model AU versus human entropy (fraction of ceiling).

encoder	pair	EU width	EU MI	AU
DINOv2-B	re-test floor	1.00 (0.61)	0.99 (0.78)	1.00
DINOv2-B	25% vs 100% data	0.96 (0.48)	0.95 (0.54)	0.97
DINOv2-B	10% vs 100% data	0.94 (0.40)	0.93 (0.42)	0.94
DINOv2-B	disjoint halves	0.94 (0.45)	0.93 (0.55)	0.95
DINOv2-B	wd $\div 10$	0.99 (0.28)	0.96 (0.19)	0.99
DINOv2-B	wd $\times 10$	0.98 (0.55)	0.98 (0.53)	0.98
DINOv2-B	AU vs human	0.36 (0.43 of $\sqrt{\text{ceiling}}$); test acc.	0.98	
CLIP-B/16	re-test floor	1.00 (0.52)	0.99 (0.73)	1.00
CLIP-B/16	25% vs 100% data	0.95 (0.33)	0.94 (0.50)	0.95
CLIP-B/16	10% vs 100% data	0.90 (0.21)	0.89 (0.39)	0.90
CLIP-B/16	disjoint halves	0.91 (0.30)	0.91 (0.52)	0.92
CLIP-B/16	wd $\div 10$	0.98 (0.11)	0.97 (0.35)	0.97
CLIP-B/16	wd $\times 10$	0.97 (0.45)	0.96 (0.58)	0.96
CLIP-B/16	AU vs human	0.43 (0.51 of $\sqrt{\text{ceiling}}$); test acc.	0.95	
ViT-B/16 sup.	re-test floor	1.00 (0.55)	0.99 (0.72)	1.00
ViT-B/16 sup.	25% vs 100% data	0.95 (0.43)	0.95 (0.49)	0.96
ViT-B/16 sup.	10% vs 100% data	0.93 (0.25)	0.92 (0.30)	0.93
ViT-B/16 sup.	disjoint halves	0.94 (0.39)	0.93 (0.48)	0.95
ViT-B/16 sup.	wd $\div 10$	0.99 (0.28)	0.97 (0.21)	0.99
ViT-B/16 sup.	wd $\times 10$	0.98 (0.43)	0.97 (0.48)	0.98
ViT-B/16 sup.	AU vs human	0.35 (0.42 of $\sqrt{\text{ceiling}}$); test acc.	0.98	
ConvNeXt-S	re-test floor	1.00 (0.55)	0.99 (0.78)	1.00
ConvNeXt-S	25% vs 100% data	0.93 (0.39)	0.93 (0.54)	0.94
ConvNeXt-S	10% vs 100% data	0.88 (0.28)	0.88 (0.41)	0.89
ConvNeXt-S	disjoint halves	0.88 (0.40)	0.88 (0.59)	0.89
ConvNeXt-S	wd $\div 10$	0.98 (0.12)	0.97 (0.35)	0.98
ConvNeXt-S	wd $\times 10$	0.97 (0.50)	0.97 (0.63)	0.97
ConvNeXt-S	AU vs human	0.34 (0.40 of $\sqrt{\text{ceiling}}$); test acc.	0.97	
MAE-B/16	re-test floor	0.99 (0.58)	0.99 (0.76)	1.00
MAE-B/16	25% vs 100% data	0.90 (0.19)	0.90 (0.41)	0.91
MAE-B/16	10% vs 100% data	0.84 (0.17)	0.84 (0.35)	0.85
MAE-B/16	disjoint halves	0.88 (0.24)	0.87 (0.48)	0.89
MAE-B/16	wd $\div 10$	0.99 (0.44)	0.98 (0.68)	0.99
MAE-B/16	wd $\times 10$	0.97 (0.52)	0.97 (0.66)	0.98
MAE-B/16	AU vs human	0.38 (0.46 of $\sqrt{\text{ceiling}}$); test acc.	0.90	

Table 7: Across 30 encoder pairs: Spearman correlation between alignment indices and item-level agreement of EU (and AU), all columns.

index	width	width partial	MI	BLR	AU
CKA	0.66	0.64	0.67	0.76	0.68
mutual k -NN	0.85	0.72	0.86	0.90	0.86
linear predictivity	0.41	0.83	0.44	0.56	0.53
$S_{\rho \rightarrow 0}$	0.88	0.83	0.90	0.93	0.93
S_{ρ} (head's ρ)	0.87	0.83	0.89	0.92	0.92